Rank	Cosine Similarity	ID	Explanation
3	0.52	16424	mathematical expressions involving polynomials and their relationships
7	0.33	14276	numerical representations and expressions related to geometry and algebra
5	0.30	17993	mathematical terms and geometric relationships
12	0.28	7528	mathematical operations and calculations
10	0.27	28227	mathematical terminology related to rational numbers and quadratic functions
7	0.26	16424	mathematical expressions involving polynomials and their relationships
6	0.26	24713	instances of mathematical notation and calculations
9	0.25	23306	mathematical equations and expressions involving variables and constants
7	0.24	11330	mathematical expressions and calculations, related to geometry and algebra
5	0.24	30063	geometric concepts and relationships within the context of triangle properties

Table 4: The top-10 most similar pairs of eigenvector-feature vector by SAE for `DeepSeek-Llama3-8B`.

Rank	Cosine Similarity	ID	Explanation
2	0.48	12085	references to scientific concepts, particularly regarding cellular and biology
1	0.36	5012	words related to the concept of "offer" and its variations
7	0.33	8927	phrases related to mathematical combinations and choices
9	0.29	10568	terms related to fractions and their properties, particularly common denominators
6	0.24	4342	mathematical expressions involving polynomials and their relationships
22	0.23	13526	mathematical expressions and equations
2	0.22	13724	instances of the word "fer" and variations or adjacent markers of some form of linguistic or structural significance
5	0.22	11268	mathematical expressions and equations
20	0.21	7675	numerical values and mathematical expressions
2	0.21	10678	instances of tab-separated values or numerical data

Table 5: The top-10 most similar pairs of eigenvector-feature vector by SAE for `Gemma-2-9b-it`.

Figure 13: Eigenvector steering visualization for feature idx 13526(mathematical concepts related to factors and multiples) for `Gemma-2-9b-it`.

spired by Wang et al. (2023); Yan et al. (2025), we leverage the gradient of the Negative Log-Likelihood (NLL) loss function to each token's hidden state at the current layer as the weight to recalculate the SAE activation score as follows:

$$\alpha^x_{grad} = \frac{1}{\|\mathcal{X}\|} \sum_{i \in \mathcal{X}} \left\| \frac{\partial L_{NLL}}{\partial r^x_i} \right\| SAE(r^x_i)$$
$$\alpha^y_{grad} = \frac{1}{\|\mathcal{Y}\|} \sum_{j \in \mathcal{Y}} \left\| \frac{\partial L_{NLL}}{\partial r^y_j} \right\| SAE(r^y_j) \tag{11}$$

where $\|\frac{\partial L_{NLL}}{\partial r^x_i}\|$ and $\|\frac{\partial L_{NLL}}{\partial r^y_j}\|$ approximately represent the importance of the current position for the model prediction. To make SAE activations α^q demonstrate sufficient robustness, the ranking result should be approximately the same as α^q_{grad}. Therefore, we employ Spearman's rank correlation coefficient (Myers et al., 2013) to measure their correlation. The result is shown in the figure 14, where we use `Gemma-2-9b-it` as the base LLM and `gemma-scope-9b-it-res(layer_9/width_`

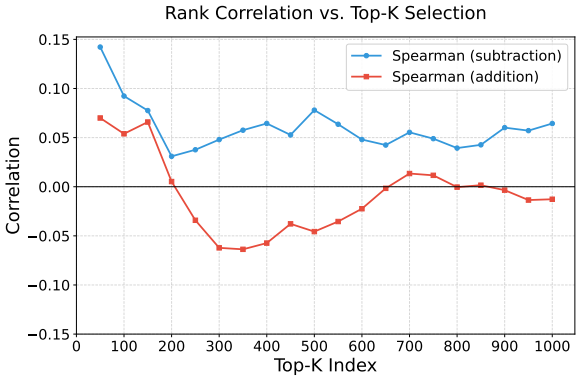

Figure 14: Spearman Correlation of Subtraction and Addition

16k/canonical) as the corresponding SAE. The result shows that compared with subtraction, addition will make the correlation lower, even less than 0, which means that the activation result obtained by subtracting can better reflect the importance of model prediction.

L Steering Case

We provide some SAE-based steering samples.

Question:Answer the following question step by step in short and put the final answer in □. Question:Carlos Montado was born on Saturday, November 9, 2002. On what day of the week will Carlos be 706 days old? (Spell out the entire day of the week.) Answer:Friday
Orign:Step 1: First, we need to calculate how many weeks are there in 706 days. To find this out, we divide 706 by 7 since each week has 7 days. Step 2: Determine if any extra days remain after subtracting complete weeks from the total number of days. After calculating the quotient and remainder when dividing 706 by 7, we'll see that there's a specific amount left over which indicates additional days beyond full weeks. Step 3: Add up all these remaining days together along with whatever fraction of a week they represent as an equivalent number of whole days or fractions thereof within one particular seven-day cycle starting at Sunday, add them onto Sunday itself. The final answer is: $\boxed{Saturday}$
Steering: Step 1: To determine the day of the week when Carlos will be 706 days old, we need to calculate how many complete weeks have passed since his birth. Step 2: We know that there are 7 days in a week, so if we divide 706 by 7, we can find the number of complete weeks. Step 3: The quotient obtained from this division is not relevant for our calculation; rather, it's essential to focus on the remainder after dividing 706 by 7. Step 4: Determine the remaining days beyond full weeks We perform the division as follows: $706 = 100 \cdot 7 + 6$. This means that there were exactly 100 complete seven-day cycles between the dates corresponding to these two numbers, with an extra six days left over. The final answer is: $\boxed{Saturday}$ Step 5: Find the day of the week based on the additional days Since there are 6 days leftover after accounting for complete weeks, we move forward 6 days from Carlos' birthday, which falls on a Saturday. Step 6: Identify the resulting day of the week Moving forward 6 days from a Saturday lands us on Friday. However, considering the instruction that requires spelling out the entire name of the day, we should present the correct response accordingly. The final answer is: $\boxed{Friday}$

Table 6: Steering Case. LLM: Llama3.1-8B-it Featue:14276(numerical representations and mathematical expressions related to geometry and algebra)

Question:Determine the modulo 4 remainder of the following sum: 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12. Answer:2
Orign: Step 1 We need to calculate $1+2+...+12$. Step 2 This is an arithmetic series with first term $a=1$, last term $l=12$, and number of terms n = 12. The formula for the sum of this type of sequence is given by$\frac{n(a+l)}{2}$, so we can use it here. Step 3 Substituting in our values gives us a total sum of $(\frac{12*13}{2})$ Step 4 Evaluating that expression yields (156) as the result. Step 5 Now we must find its remainder when divided by four using modular arithmetic (or simply division). Step 6 To do this, divide 156 by 4. Step 7 When you perform integer division between these two numbers, you get quotient q = 39 and remainder r = 0. Step 8 Since there's no remainder left after dividing, then your final answer will be just the value of the remainder which is zero. I hope it is correct. Thank you!
Steering: Step 1: We need to find the modulo 4 remainder for each number in the given sequence. To do this, we will divide each number by 4 and take note of its remainder. Step 2: Determine the remainders when dividing numbers from 1 through 12 by 4. The remainders are as follows: - $1 \div 4 = 0$ with a remainder of 1 - $2 \div 4 = 0$ with a remainder of 2 - $3 \div 4 = 0$ with a remainder of 3 - $4 \div 4 = 1$ with a remainder of 0 - $5 \div 4 = 1$ with a remainder of 1 - $6 \div 4 = 1$ with a remainder of 2 - $7 \div 4 = 1$ with a remainder of 3 - $8 \div 4 = 2$ with a remainder of 0 - $9 \div 4 = 2$ with a remainder of 1 - $10 \div 4 = 2$ with a remainder of 2 - $11 \div 4 = 2$ with a remainder of 3 - $12 \div 4 = 3$ with a remainder of 0 Step 3: Group the numbers based on their respective remainders (0, 1, 2, 3) since these groups can be simplified using modular arithmetic properties. We notice that there is one group for each possible remainder mod 4: - Remainder 0: $\{4, 8, 12\}$ - Remainder 1: $\{1, 5, 9\}$ - Remainder 2: $\{2, 6, 10\}$ - Remainder 3: $\{3, 7, 11\}$ Step 4: Sum up all the elements within each group because they have equivalent values under modulo 4 addition. Each set sums up to - For remainder 0: $4+8+12=24$, which has a remainder of 0 after division by 4, - For remainder 1: $1+5+9=15$, which has a remainder of 3 after division by 4, - For remainder 2: $2+6+10=18$, which has a remainder of 2 after division by 4, - For remainder 3: $3+7+11=21$, which has a remainder of 1 after division by 4, Step 5: Calculate the total sum of the grouped sets' remainders to determine the overall modulo 4 remainder of the original series. Adding together the individual results yields $(0)+(3)+(2)+(1)=6$. Then taking the result modulo 4 gives us: $$6 \mod 4 = 2.$$ The final answer is: $\boxed{2}$

Table 7: Steering Case. LLM: Llama3.1-8B-it Featue:24715(logical conclusions or summarizing statements in procedural or mathematical text)